

CDACL-006

MARKET ANALYSIS

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Executive summary

This project analyses the customer purchasing patterns and the products performance using the grocery market data. We have used **SQL and Power BI** to extract the insights from the orders, products and the department tables and to visualize the business trends. Also we have verifies the results in both the tools. Through this analysis we identified the high-performing products, reorder pattern, customer purchase pattern and peak shopping periods. Based on the findings, recommendations were developed to improve the inventory management, marketing strategies, customer retention and the overall business performance.

Project Objective

The main goal of the project is to analyze the customer buying behavior and product performance based on the transactional market order data. The project aims to extract meaningful insights from customer orders, products, departments and aisles to support data-driven business decision making. This will be achieved by implementing SQL and PowerBI techniques on a relational database.

The research aims to find buying habits, best products, reordering, department wise performance, customer ordering patterns and shopping habits. The project uses data cleaning, relational table joins, aggregation and analytical queries to turn raw transactional data into actionable business insights.

The project also aims to assess consumer preferences based on popular products, peak ordering times, order frequency, average basket size and product reorder rates. These findings contribute to the understanding of customer loyalty and purchasing behavior in various product categories.

The project, following the analysis results, suggests business recommendations to optimize marketing strategies, improve inventory management, improve product placement, strengthen customer retention and increase overall customer satisfaction. The insights generated can help in better business planning, personalized marketing campaigns, demand forecasting and operational decision-making for retail and e-commerce organizations.

To put it simply, this project shows how data analysis can serve to translate huge amounts of transactional data into a rich business intelligence to help make strategic choices and to improve organizational performance.

Dataset Overview

The dataset used in this project has transactional market orders data and has five tables. The tables are connected through primary keys and foreign keys helping in comprehensive analysis of customer purchase behaviour and the product performance.

Orders

This table stores customer order information, including order ID, user ID, order sequence, order day, order hour, and days since the previous order. It is used to analyse customer ordering patterns and shopping trends.

Order Products

This table stores the products purchased in each order, along with the sequence in which they were added to the cart and their reorder status. This table is the primary source for analysing product purchases and reorder behaviour.

Products

This table contains product details such as product name, aisle ID, and department ID. This table was cleaned before analysis to ensure data accuracy and consistency.

Aisles

This table contains all product aisles in the store and is used for aisle-wise product analysis.

Departments

This table stores product department information, enabling department-level analysis of customer purchases and product performance.

Table Relationships

The tables are connected as follows:

- **Orders** → **Order Products** using **Order ID**
- **Order Products** → **Products** using **Product ID**
- **Products** → **Aisles** using **Aisle ID**
- **Products** → **Departments** using **Department ID**

These relationships allow data from multiple tables to be combined using joins to generate meaningful business insights and support customer behavior analysis.

Data Cleaning and Preprocessing

The dataset was examined to find data quality problems that would compromise the correctness of the findings before the SQL analysis was carried out. The Products table was exported for preprocessing since the project's source database was read-only.

A few records with overlapping product information were discovered during the data review, leading to inconsistent product details and missing Product IDs. In order to separate the combined data, restore the missing product information, and guarantee that each row represented a single legitimate product with the correct Product ID, Product Name, Aisle ID, and Department ID, these records were located and fixed using Excel formulas.

Following the cleaning process, the dataset was checked for missing values, duplicate Product IDs, and accurate linkages between goods, aisles, and departments. The cleaned Products table was then imported into the MySQL database, where row counts and SQL queries were used to ensure data integrity and then to Power BI for visualization.

These pretreatment methods increased the dataset's quality and consistency, allowing for accurate analysis and dependable business insights.

Methodology

We followed a structured data analysis approach to examine customer purchasing behavior and product performance using a cleaned retail dataset. The methodology consisted of the following stages:

1. Data Collection

The dataset containing customer orders, products, departments, aisles, and order-product transactions was obtained from the project database.

2. Data Cleaning and Preprocessing

The dataset was reviewed to identify data quality issues. Since the source database was read-only, the Products table was exported and cleaned separately. Overlapping product information and missing product details were corrected using Excel formulas. The cleaned table was then imported into a MySQL database and Power BI and validated before analysis.

3. Database Preparation

The cleaned table along with other tables was imported to MySQL and PowerBI. Relationships between the tables were verified using their primary and foreign keys to ensure data consistency and accurate analysis.

4. Data Analysis

The cleaned dataset was analysed using two complementary tools:

- **MySQL** was used to perform analytical queries involving joins, aggregations, subqueries, Common Table Expressions (CTEs), and window functions to answer the project questions.
- **Power BI** was used to independently explore the cleaned dataset through interactive visualizations, enabling the identification of customer purchasing patterns, product performance, and business trends.

5. Result Validation

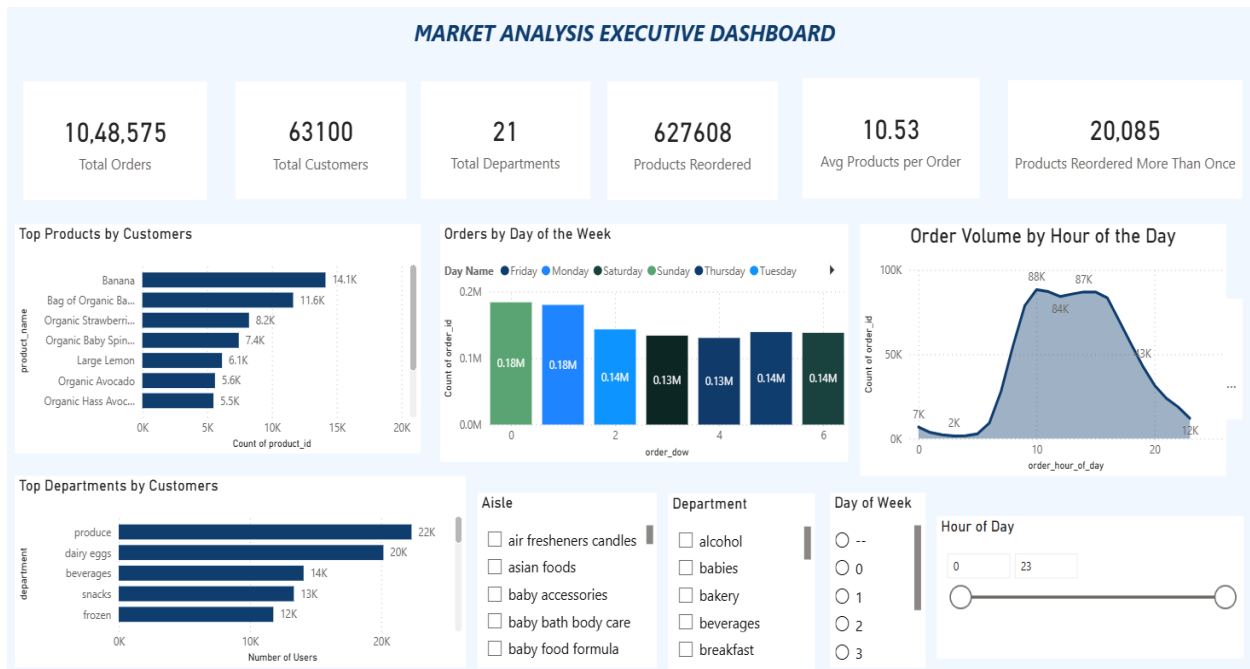
The analytical results were validated by verifying row counts, checking table relationships, and cross-checking outputs in both MySQL and Power BI to ensure consistency and accuracy across the analyses.

6. Business Insights and Recommendations

The findings from the SQL analysis and Power BI visualizations were interpreted to understand customer purchasing behavior and product performance. Based on these insights, recommendations were developed to optimize marketing strategies, improve inventory management, and enhance customer satisfaction.

Project summary dashboard

KPI	Value
Total Orders	1,048,575
Total Products	104,879
Total Departments	21
Total Aisles	134
Unique Customers	63,100
Average Products per Order	10.53
Most Ordered Product	Banana
Highest Customer Department	Produce

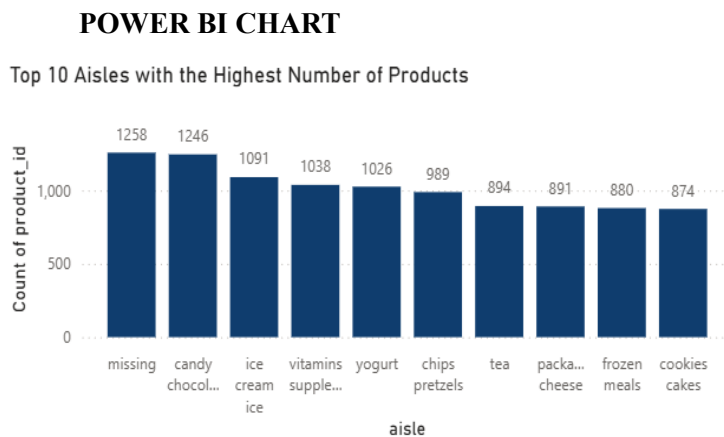


Analysis and insights

1. What are the top 10 aisles with the highest number of products?

SQL QUERY

```
SELECT
a.aisle_id,a.aisle,
COUNT(p.product_id)
AS product_count
FROM aisles a
JOIN products p
ON a.aisle_id =
p.aisle_id
GROUP BY a.aisle_id,
a.aisle
ORDER BY product_count
DESC
LIMIT 10;
```



Analysis

The analysis showed the aisle named **Missing** containing the highest number of products (1,258), followed by **Candy & Chocolate** (1,246), **Ice Cream & Ice** (1,091), **Vitamins & Supplements** (1,038), and **Yogurt** (1,026). Several other aisles, including **Chips & Pretzels**, **Tea**, and **Packaged Cheese**, also contained a large number of products, indicating a broad assortment across popular grocery categories.

The results show that product offerings are concentrated in a few major aisles, particularly snacks, frozen foods, dairy products, and health-related items. The presence of the **Missing**

aisle represents incomplete product classification in the original dataset rather than an actual business category and should therefore be interpreted as a data quality issue rather than a retail trend.

2. How many unique departments are there in the dataset?

SQL QUERY

```
SELECT COUNT(DISTINCT department_id)
AS unique_departments
FROM departments;
```

POWER BI CHART

Unique Departments

21

Total Departments

Analysis

The dataset contains **21 unique departments**, representing a wide range of grocery categories, including Produce, Dairy & Eggs, Beverages, Pantry, Snacks, Household, Personal Care, and Frozen Foods. This demonstrates that the dataset covers multiple business areas and supports comprehensive retail analysis.

A diverse departmental structure enables detailed category-level analysis, helping retailers evaluate product performance, understand customer purchasing patterns, and make informed inventory and merchandising decisions.

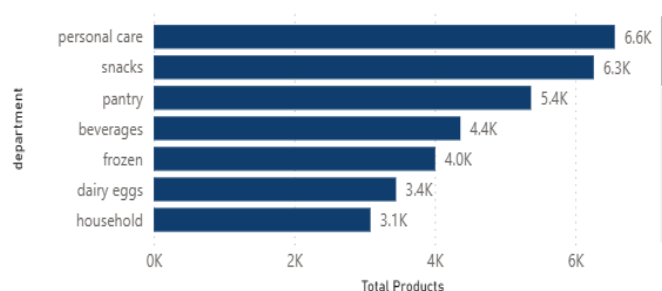
3. What is the distribution of products across departments?

SQL QUERY

```
SELECT
d.department_id,d.department,
COUNT(p.product_id)
AS product_count
FROM departments d
JOIN products p
ON d.department_id =
p.department_id
GROUP BY d.department_id,
d.department
ORDER BY product_count DESC;
```

POWER BI CHART

Distribution of Products Across Department



Analysis

The **Personal Care** department contains the highest number of products (6,563), followed by **Snacks (6,264)**, **Pantry (5,371)**, **Beverages (4,365)**, and **Frozen (4,007)**. In comparison, departments such as **Bulk**, **Other**, and **Meat & Seafood** contain considerably fewer products.

The findings indicate that product assortment is concentrated in selected departments. However, a larger product range does not necessarily indicate higher customer demand, as later analyses show that departments with fewer products, such as Produce, attract significantly more customers and purchase activity.

4. What are the top 10 products with the highest reorder rates?

SQL QUERY

```
SELECT product_id,  
       COUNT(*) AS  
total_orders,  
       SUM(reordered) AS  
reorder_count,  
       ROUND(AVG(reordered)  
* 100, 2) AS reorder_rate  
FROM order_products_train  
GROUP BY product_id  
HAVING COUNT(*) >= 10  
ORDER BY reorder_rate  
DESC, total_orders DESC  
LIMIT 10;
```

POWER BI CHART

Top 10 Products with the Highest Reorder Rates

product_name	Reorder Rate	Total_Orders	Reorder Count
Vanilla Bottles	100.00%	11	11
Total Greek Strained Yogurt with Blueberry	100.00%	16	16
Premium Apple Juice	100.00%	12	12
Organic Blueberry Cream On Top Whole Milk Yogurt	100.00%	11	11
Naked Meat Free Chik'n Cutlets	100.00%	14	14
Mocha Cappuccino	100.00%	13	13
Dark Italian Roast Ground Coffee	100.00%	14	14
Dark Chocolate Squares With 88% Cocoa	100.00%	10	10
Cottage Cheese, Natural Small Curd, 4% Milkfat Min	100.00%	10	10
Chocolate Energy Supplement	100.00%	12	12
Belvita Soft Baked Mixed Berry	100.00%	10	10
Total	100.00%	133	133

Analysis

The analysis identified ten products with a **100% reorder rate**, meaning every recorded purchase for these products was a repeat purchase. To improve the reliability of the analysis, only products with **at least 10 recorded orders** were included, reducing the influence of products with very few transactions.

A perfect reorder rate indicates exceptionally strong customer loyalty and consistent repeat purchasing. Although these products may not have the highest sales volumes, they demonstrate stable customer demand and represent valuable products for inventory planning, customer retention strategies, and personalized marketing campaigns.

5. How many unique users have placed orders in the dataset?

SQL QUERY

```
SELECT COUNT(DISTINCT user_id)  
AS unique_users  
FROM orders;
```

POWER BI CHART

63100
Total Customers

Analysis

The dataset contains **63,100 unique customers** who placed at least one order during the analysis period. This large customer base provides a reliable foundation for evaluating purchasing behaviour, reorder patterns, and product demand across multiple departments and categories.

The substantial number of active customers enhances the credibility of the analytical findings and enables meaningful business insights that can support customer segmentation, demand forecasting, and strategic decision-making.

6. What is the average number of days between orders for each user?

SQL QUERY

```
SELECT user_id,
ROUND(AVG(days_since_prior_order),
2) AS avg_days_between_orders
FROM orders
WHERE days_since_prior_order IS
NOT NULL
GROUP BY user_id
ORDER BY user_id;
```

POWER BI CHART

Average Number of Days Between Orders for Each User

user_id	Average Days Between Orders
1	19.00
2	16.29
3	12.00
4	17.00
5	11.50
6	13.33
7	10.45
8	23.33
9	22.00
10	21.80

Analysis

The analysis shows that customers place orders at varying intervals, indicating different shopping frequencies across the customer base. While some customers purchase groceries within a few days, others take longer between orders, reflecting diverse shopping habits and purchasing needs.

Understanding the average time between orders helps retailers identify purchasing cycles and forecast customer demand more accurately. These insights can support inventory planning, personalized marketing campaigns, and timely promotional offers aimed at encouraging repeat purchases.

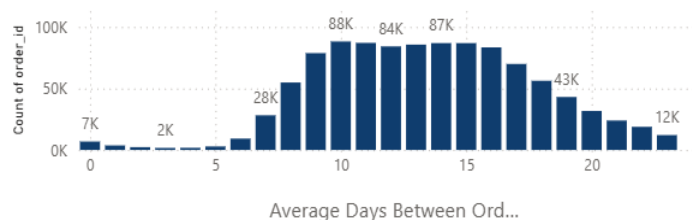
7. What are the peak hours of order placement during the day?

SQL QUERY

```
SELECT
order_hour_of_day,
COUNT(*) AS
order_count FROM orders
GROUP BY
order_hour_of_day
ORDER BY order_count
DESC;
```

POWER BI CHART

Peak Hours of Order Placement During the Day



Analysis

Customer ordering activity is highest during the **afternoon hours**, with **2:00 PM (14:00)** recording the maximum number of orders (**86,905**), followed closely by **3:00 PM** and **1:00 PM**. Order volumes remain consistently high between **12:00 PM and 4:00 PM**, while early morning hours, particularly **12:00 AM to 5:00 AM**, show the lowest level of customer activity.

The results indicate that customers prefer shopping during the afternoon, making this the most critical period for retail operations. Businesses can use these insights to optimize

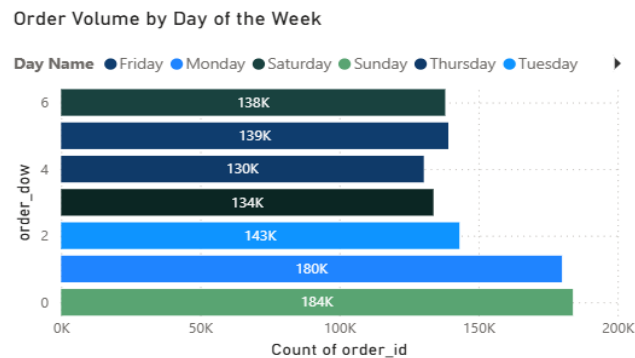
staffing, maintain adequate inventory, and schedule promotional campaigns during peak shopping hours to improve customer experience and operational efficiency.

8. How does order volume vary by day of the week?

SQL QUERY

```
SELECT order_dow,
       COUNT(*) AS order_count
FROM orders
GROUP BY order_dow
ORDER BY order_dow;
```

POWER BI CHART



Analysis

The analysis shows noticeable variation in customer ordering activity throughout the week. **Day 0** recorded the highest number of orders (**183,939**), followed by **Day 1 (180,025)**, while **Day 4** recorded the lowest order volume (**130,367**). Although customer activity remains relatively consistent across the week, certain days experience significantly higher shopping demand.

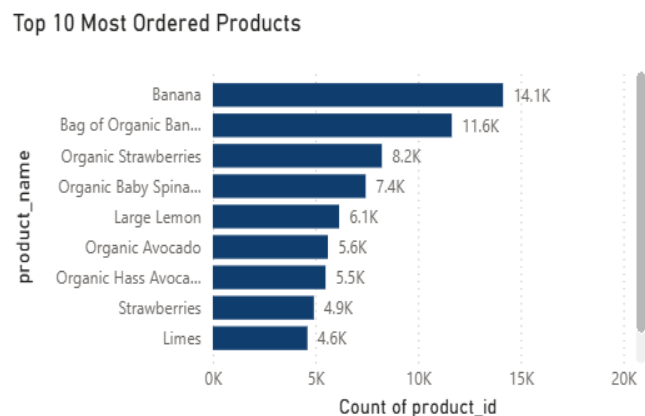
These purchasing patterns indicate that customer demand fluctuates throughout the week. Understanding these trends enables retailers to optimize staffing schedules, improve inventory planning, and launch targeted promotional campaigns during high-demand periods to maximize sales and customer satisfaction.

9. What are the top 10 most ordered products?

SQL QUERY

```
SELECT p.product_id,
       p.product_name,
       COUNT(op.order_id)
AS order_count
FROM products p
JOIN order_products_train
op ON p.product_id =
op.product_id
GROUP BY p.product_id,
p.product_name
ORDER BY order_count DESC
LIMIT 10;
```

POWER BI CHART



Analysis

The analysis identified **Banana** as the most frequently ordered product with **14,136 orders**, followed by **Bag of Organic Bananas (11,639)**, **Organic Strawberries (8,233)**, **Organic Baby Spinach (7,443)**, and **Large Lemon (6,148)**. Most of the highest-selling products belong to the **Produce** department, highlighting strong customer demand for fresh fruits and vegetables.

The findings indicate that customers consistently prioritize fresh and essential grocery products over discretionary items. Retailers should ensure these products remain well-stocked and prominently displayed, as they contribute significantly to overall sales and customer satisfaction.

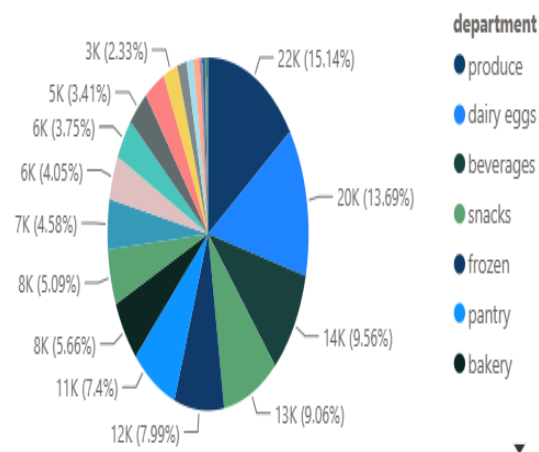
10. How many users have placed orders in each department?

SQL QUERY

```
SELECT
d.department_id,d.department,
COUNT(*) AS user_count
FROM (
    SELECT DISTINCT
o.user_id,p.department_id FROM
orders o
    JOIN order_products_train
op ON o.order_id = op.order_id
    JOIN products p ON
op.product_id = p.product_id
) AS user_departments
JOIN departments d ON
user_departments.department_id
= d.department_id
GROUP BY
d.department_id,d.department
ORDER BY user_count DESC;
```

POWER BI CHART

Number of Users by Department



Analysis

The **Produce** department recorded the highest customer engagement, with **22,317 unique customers**, followed by **Dairy & Eggs (20,174)**, **Beverages (14,090)**, **Snacks (13,346)**, and **Frozen (11,783)**. In contrast, departments such as **Bulk**, **Other**, and **Pets** attracted considerably fewer customers.

The results demonstrate that customers primarily purchase products from essential grocery departments. These departments should be prioritized for inventory management, merchandising, and promotional activities, while lower-performing departments may require targeted marketing strategies to improve customer engagement.

11. What is the average number of products per order?

SQL QUERY

POWER BI CHART

```
SELECT
ROUND(COUNT(*)/COUNT(DISTINCT
order_id),2)
AS avg_products_per_order
from order_products_train;
```

10.53
Avg Products per Order

Analysis

The analysis shows that customers purchase an average of **10.53 products per order**, indicating that shoppers generally buy multiple products during each shopping trip rather than purchasing individual items. This reflects typical grocery shopping behavior, where customers tend to purchase a variety of products in a single transaction.

The average basket size provides valuable insights for retailers when designing product bundles, cross-selling strategies, and promotional offers. Encouraging customers to add complementary products to their baskets can further increase the average order value and overall sales.

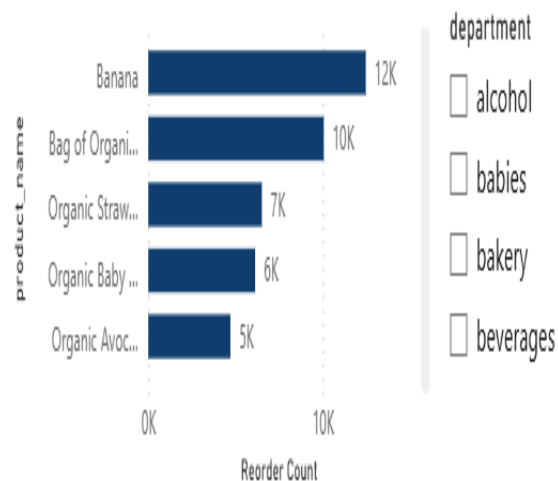
12. What are the most reordered products in each department?

SQL QUERY

```
WITH product_reorders
AS (SELECT
d.department_id,d.department,
p.product_id,p.product_name,
SUM(op.reordered) AS
reorder_count
FROM order_products_train op
JOIN products p
ON op.product_id =
p.product_id
JOIN departments d
ON p.department_id =
d.department_id
GROUP BY
d.department_id,d.department,
p.product_id,p.product_name
),
ranked_products AS (SELECT *,
DENSE_RANK() OVER (
PARTITION BY
department_id
ORDER BY
reorder_count DESC) AS
product_rank
FROM product_reorders)
SELECT department_id,
department,product_id,
product_name,reorder_count
FROM ranked_products
WHERE product_rank = 1
ORDER BY department_id;
```

POWER BI CHART

Most Reordered Products by Department



Analysis

The analysis identified the highest reordered product within each department. Products such as **Banana** (Produce), **Organic Whole Milk** (Dairy & Eggs), **Sparkling Water Grapefruit** (Beverages), **Original Hummus** (Deli), and **Boneless Skinless Chicken Breasts** (Meat & Seafood) recorded the highest reorder counts in their respective departments. These products consistently demonstrate strong customer preference and repeat purchasing behavior.

The findings indicate that essential grocery items dominate reorder activity across multiple departments. Retailers should prioritize the availability of these products through effective inventory management and consider featuring them in promotional campaigns to strengthen customer loyalty and maximize repeat sales.

13. How many products have been reordered more than once?

SQL QUERY

```
SELECT COUNT(*) AS  
products_reordered_more_than_once  
FROM (  
    SELECT product_id  
    FROM order_products_train  
    GROUP BY product_id  
    HAVING SUM(reordered) > 1  
) AS reordered_products;
```

POWER BI CHART

20,085

Products Reordered More
Than Once

Analysis

The analysis identified **20,085 products** that were reordered more than once, indicating that a substantial proportion of the product catalog experiences repeat purchases. This demonstrates that customer purchasing behavior extends beyond one-time transactions and reflects ongoing demand for a wide range of products.

A high number of reordered products suggests strong customer satisfaction and consistent purchasing habits. These insights can support inventory optimization, demand forecasting, and customer retention strategies by identifying products that contribute to sustained repeat business.

14. What is the average number of products added to the cart per order?

SQL QUERY

```
SELECT ROUND(AVG(product_count), 2)  
AS avg_products_per_order  
FROM (SELECT order_id,  
    COUNT(*) AS product_count  
    FROM order_products_train  
    GROUP BY order_id  
) AS order_sizes;
```

POWER BI CHART

10.53

Average Products Per Order

Analysis

The analysis shows that customers add an average of **10.53 products** to their shopping cart per order. This result is consistent with the average basket size observed in the previous analysis, confirming that customers typically purchase multiple products during each shopping session.

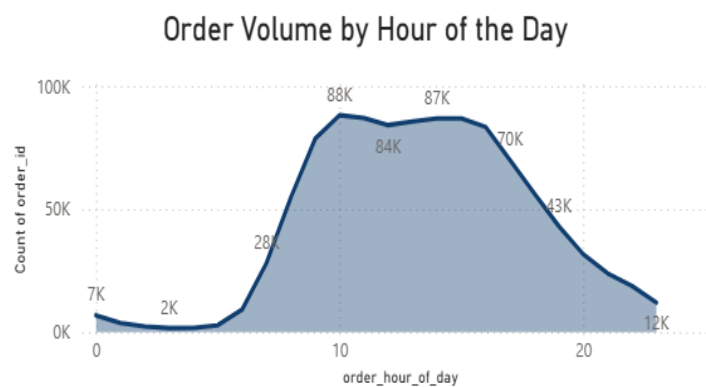
The consistency in average basket size highlights opportunities for retailers to implement cross-selling and product bundling strategies. Recommending complementary products during checkout can further increase basket value while improving the overall customer shopping experience.

15. How does the number of orders vary by hour of the day?

SQL QUERY

```
SELECT
order_hour_of_day,
COUNT(DISTINCT
order_id) AS
total_orders
FROM orders
GROUP BY
order_hour_of_day
ORDER BY
order_hour_of_day;
```

POWER BI CHART



Analysis

Customer order activity varies considerably throughout the day. Order volumes gradually increase during the morning hours, reach their highest levels between **10:00 AM and 4:00 PM**, and then decline during the evening and overnight hours. The highest order activity occurs during the afternoon, while the lowest activity is observed during the early morning hours.

The distribution indicates that customer shopping follows a predictable daily pattern. Retailers can use these insights to optimize staffing schedules, ensure adequate stock availability during busy periods, and schedule promotional campaigns when customer engagement is highest.

16. What is the distribution of order sizes (number of products per order)?

SQL QUERY

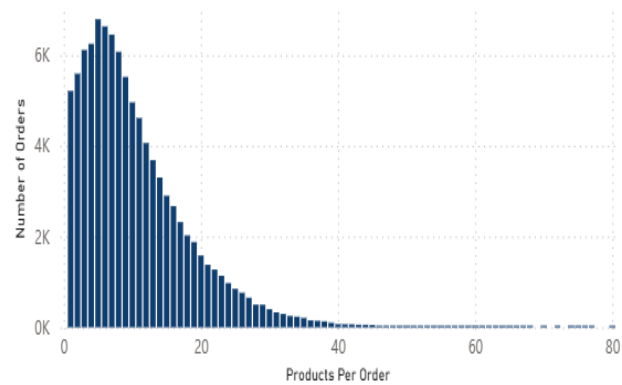
POWER BI CHART

```

WITH order_sizes AS (
SELECT order_id,
       COUNT(*) AS
products_per_order
  FROM order_products_train
  GROUP BY order_id)
SELECT products_per_order
AS order_size,
       COUNT(*) AS
number_of_orders,
       ROUND(100.0 * COUNT(*)/
SUM(COUNT(*) OVER ( ),2)
AS percentage_of_orders
  FROM order_sizes
  GROUP BY products_per_order
  ORDER BY products_per_order;

```

Number of Orders by Products Per Order



Analysis

The analysis shows that **small and medium-sized orders are the most common** among customers. Orders containing **4 to 8 products** account for the largest share of transactions, with **5-product orders (6.82%)** representing the highest proportion. As the number of products per order increases beyond this range, the percentage of orders gradually decreases, and orders containing more than 30 products become relatively uncommon.

The findings indicate that customers generally purchase a moderate number of products during each shopping trip rather than placing very large orders. Understanding order size distribution enables retailers to design effective product bundles, optimize promotional offers, and improve inventory planning based on typical customer purchasing behaviour.

17. What is the average reorder rate for products in each aisle?

SQL QUERY

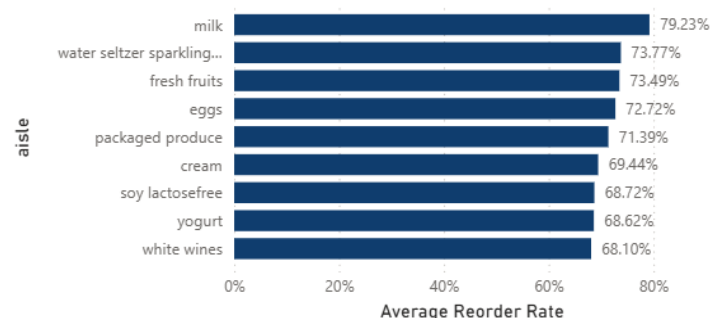
```

SELECT
a.aisle, ROUND(AVG(op.reorder
d),3) AS average_reorder_rate
  FROM order_products_train op
  JOIN products p
  ON op.product_id =
p.product_id
  JOIN aisles a
  ON p.aisle_id = a.aisle_id
  GROUP BY a.aisle
  ORDER BY average_reorder_rate
  DESC;

```

POWER BI CHART

Average Reorder Rate by aisle



Analysis

The analysis shows that aisles containing **everyday grocery essentials** record the highest reorder rates. **Milk (79.2%)**, **Water, Seltzer & Sparkling Water (73.8%)**, **Fresh Fruits (73.5%)**, **Eggs (72.7%)**, and **Packaged Produce (71.4%)** are among the highest-performing aisles, indicating that customers repeatedly purchase these products over time.

The consistently high reorder rates demonstrate strong customer loyalty toward essential grocery items. Retailers should prioritize these categories for inventory management, shelf availability, and targeted promotional campaigns to ensure continued customer satisfaction and maximize repeat sales.

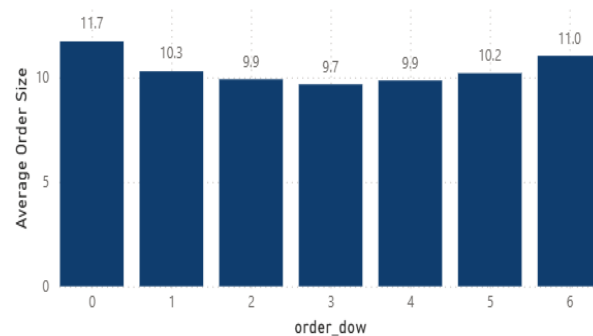
18. How does the average order size vary by day of the week?

SQL QUERY

```
SELECT o.order_dow
AS day_of_week,
COUNT(DISTINCT o.order_id)
AS total_orders,
ROUND(AVG(s.order_size), 2)
AS average_order_size
FROM (SELECT order_id,
COUNT(product_id)
AS order_size
FROM order_products_train
GROUP BY order_id
) AS s JOIN orders o
ON s.order_id = o.order_id
GROUP BY o.order_dow
ORDER BY o.order_dow;
```

POWER BI CHART

Average Order Size by Day of the Week



Analysis

The analysis indicates that **Day 0** has the highest average order size (**11.72 products per order**), followed by **Day 6 (11.03)** and **Day 1 (10.29)**. The remaining days show relatively similar average basket sizes, ranging between approximately **9.67 and 10.20 products**.

The variation in average order size suggests that customers tend to purchase more products on certain days of the week. These insights can assist retailers in forecasting inventory requirements, allocating staff efficiently, and planning promotional activities during periods when customers are likely to make larger purchases.

19. What are the top 10 users with the highest number of orders?

SQL QUERY

```
SELECT user_id,
COUNT(DISTINCT order_id)
AS total_orders
FROM orders
GROUP BY user_id
ORDER BY total_orders DESC
LIMIT 10;
```

POWER BI CHART

Top 10 Users with the Highest Number of Orders

user_id	Count of order_id
210	100
310	100
313	100
690	100
786	100
964	100
1024	100
1310	100
1420	100
1428	100

Analysis

The analysis identified the **top 10 customers** based on the total number of orders placed. Each of these customers completed **100 orders**, demonstrating exceptionally high purchasing frequency compared to the rest of the customer base.

The presence of highly active customers indicates a valuable segment of loyal shoppers who contribute significantly to repeat business. Retailers can strengthen customer retention by offering personalized rewards, exclusive promotions, and loyalty program benefits to encourage continued engagement and long-term customer value.

20. How many products belong to each aisle and department?

SQL QUERY

```
SELECT a.aisle_id,a.aisle,
d.department_id,d.department,
COUNT(DISTINCT p.product_id)
AS product_count
FROM products p JOIN aisles a
ON p.aisle_id = a.aisle_id
JOIN departments d
ON p.department_id =d.department_id
GROUP BY a.aisle_id,a.aisle,
d.department_id,d.department
ORDER BY d.department_id,
product_count DESC;
```

POWER BI CHART

Number of Products by Aisle and Department



Analysis

The analysis reveals considerable variation in product distribution across aisles and departments. The **Personal Care** department contains the largest assortment of products, with major aisles such as **Vitamins & Supplements**, **Hair Care**, and **Oral Hygiene** contributing substantially to its product range. Similarly, the **Snacks**, **Frozen**, and **Pantry** departments also contain a broad selection of products, while departments such as **Bulk** and **Other** have comparatively smaller assortments.

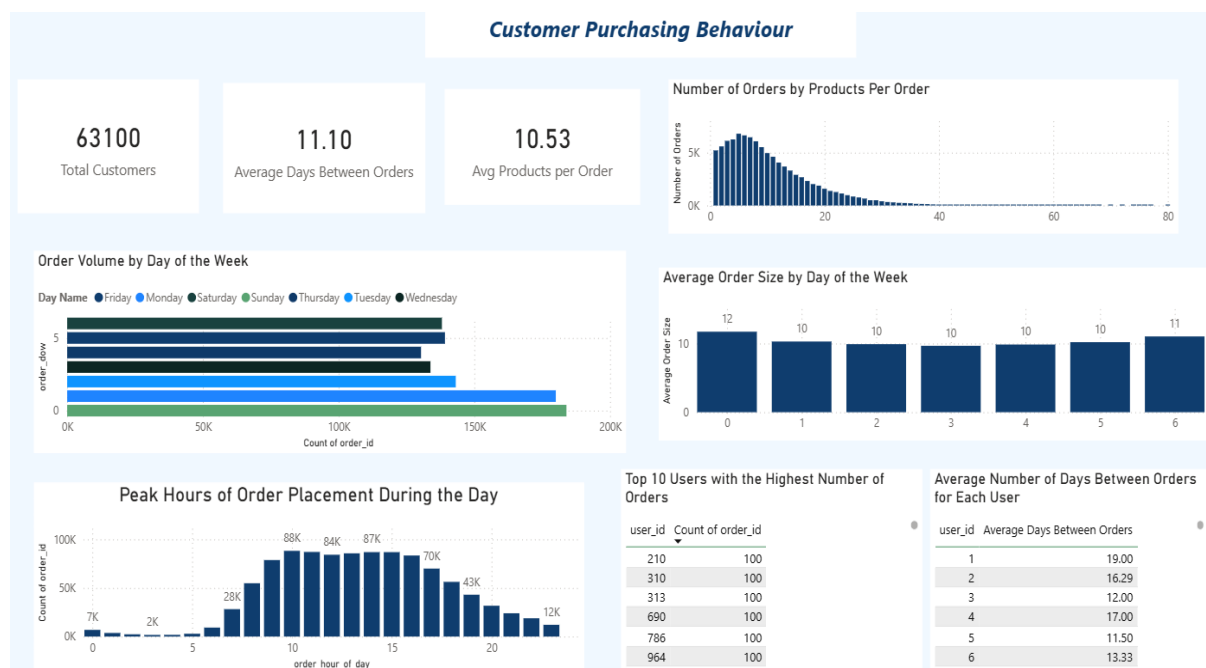
The findings indicate that product assortment is not directly related to customer demand. Earlier analyses showed that **Produce** attracts the highest number of customers and contains many of the most frequently purchased products despite offering fewer product categories than **Personal Care**. This highlights the importance of balancing product variety with customer purchasing behaviour when making merchandising, inventory, and product assortment decisions.

Customer Purchasing Behavior Analysis

We evaluated customer purchasing behavior using the queries in order to determine shopping trends, ordering frequency, customer preferences, and repeat purchase behavior. Several significant trends that offer insightful information about consumer purchasing patterns were identified by the investigation.

According to the findings, consumers mostly buy necessities like fresh produce, dairy, and drinks. **Organic strawberries, bananas, bags of organic bananas, and organic whole milk** were among the most often requested and reordered items, displaying a strong consumer preference for daily essentials. Fresh food categories are the main drivers of client purchases, as evidenced by the Produce department's biggest number of unique customers.

Customer shopping activity was not evenly distributed throughout the day. The majority of orders were placed during the **afternoon hours**, with **2:00 PM** recording the highest order volume. Order activity remained high between **12:00 PM and 4:00 PM**, while early morning hours had the lowest number of orders. This suggests that customers prefer shopping during the afternoon, making it an ideal period for inventory availability, staffing, and promotional campaigns.



Weekly purchasing patterns also showed noticeable variations. **Day 0** recorded both the **highest number of orders** and the **largest average order size**, while mid-week days mostly had relatively lower customer activity. This indicates that customers tend to make larger grocery purchases at specific times during the week rather than distributing their shopping evenly across all days.

The analysis further revealed strong repeat purchasing behavior. More than **20,000 products** were reordered more than once, and several product categories such as **Milk, Fresh Fruits, Eggs, Water, and Packaged Produce** achieved high reorder rates. These findings indicate that customers regularly repurchase essential grocery items, demonstrating consistent buying habits and strong product loyalty.

Order size analysis showed that customers purchased an average of **10.53 products per order**, while the most common order sizes ranged between **4 and 7 products**. This suggests that although customers typically purchase multiple products during each shopping trip, extremely large orders are comparatively less common.

Finally, identifying consumers with **100 completed orders** indicates the presence of a highly devoted customer base. These recurrent customers contribute significantly to overall sales and provide tremendous chances for customer retention via targeted offers and loyalty programs.

Overall, the purchasing behavior research shows that customers engage in planned grocery shopping activity, as seen by regular purchases of important products, frequent repeat purchases, modest basket sizes, and predictable shopping patterns across different times of day and week.

Product Performance Analysis

Product performance was evaluated by analyzing product popularity, reorder behavior, departmental performance, and product distribution. The findings provide valuable insights into which products contribute most to customer demand and overall business performance.

The analysis identified **Banana** as the highest-performing product, recording the highest number of customer orders and also the highest reorder count within the Produce department. Other products such as **Bag of Organic Bananas, Organic Strawberries, Organic Baby Spinach**, and **Organic Whole Milk** also showed strong sales performance, indicating a consistency in the customer preference for fresh and organic grocery items.

Department-level analysis showed that **Produce** is the strongest-performing department, attracting the highest number of unique customers. **Dairy & Eggs, Beverages**, and **Snacks** also recorded high customer engagement, demonstrating that they were present in customers regular shopping baskets. These departments primarily consist of essential grocery products that customers purchase frequently.



Reorder analysis further highlighted strong product performance across essential grocery categories. Products such as **Bananas**, **Organic Whole Milk**, **Original Hummus**, and **Sparkling Water Grapefruit** recorded the highest reorder counts within their respective departments. In addition, aisles including **Milk**, **Fresh Fruits**, **Eggs**, and **Water, Seltzer & Sparkling Water** had the highest reorder rates, indicating strong customer satisfaction and repeat purchasing behavior.

The analysis also revealed that **product assortment does not always correspond to customer demand**. Although the **Personal Care** department contains the largest number of products, it does not attract the highest number of customers. In contrast, the **Produce** department, with a comparatively smaller assortment, consistently generated higher customer engagement, more frequent purchases, and stronger reorder performance. This demonstrates that customer demand is influenced more by product necessity and purchasing frequency than by the number of products available.

Overall, the product performance analysis indicates that **fresh produce, dairy products, beverages, and other everyday grocery essentials** are the primary contributors to customer demand and repeat purchases. These high-performing products should be prioritized for inventory management, stock replenishment, merchandising, and promotional campaigns to maximize sales and improve customer satisfaction.

Key Findings

- **Banana** is the best performing product based on both order volume and repeat purchases.
- **Produce** is the most successful department in terms of customer engagement.

- Essential grocery categories such as **Produce, Dairy & Eggs, and Beverages** consistently outperform other departments.
- High reorder rates indicate **strong customer satisfaction and loyalty** for frequently purchased grocery essentials.
- A larger product assortment does not necessarily result in higher customer demand, emphasizing the importance of stocking products that customers purchase regularly.

Marketing Strategy Recommendations

Based on the analysis of customer purchasing behavior and product performance, the following strategies are recommended to improve customer satisfaction, increase sales, and enhance overall business performance.

1. Prioritize High-Demand Products

The most reordered items, according to the data, were **bananas, bags of organic bananas, organic whole milk, and organic strawberries**. To avoid stockouts and maintain customer happiness, retailers should guarantee that these products are consistently available by using effective inventory management and prompt stock replenishment.

2. Develop Personalized Promotions

Strong recurring purchasing behavior for necessary grocery goods is indicated by customer shopping patterns. Personalized product recommendations, digital coupons, and targeted discounts on commonly purchased items can be offered by the retailer using purchase history. Personalized marketing can boost repeat business and increase consumer engagement.

3. Strengthen Customer Loyalty Programs

The presence of customers with a high number of completed orders demonstrates strong customer loyalty. Implementing reward points, exclusive discounts, cashback offers, or membership benefits for frequent shoppers can improve customer retention and increase long-term customer value.

4. Optimize Inventory Planning

Peak shopping activity occurs during the afternoon hours, while Produce, Dairy & Eggs, and Beverages consistently experience the highest customer demand. Retailer should maintain adequate stock levels for these departments, particularly during high-demand periods, to improve product availability and reduce lost sales opportunities.

5. Improve Product Visibility

Products with consistently high reorder rates should receive greater visibility through featured displays, promotional banners, and strategic shelf placement. Highlighting best-

selling and frequently purchased products can encourage additional purchases and enhance the customer shopping experience.

6. Enhance Cross-Selling and Product Bundling

Customers purchase an average of more than **10 products per order**, indicating opportunities for cross-selling. Retailers can increase basket value by offering bundled promotions, such as pairing fresh produce with dairy products, beverages, or bakery items that are commonly purchased together.

7. Optimize Low-Performing Categories

Departments such as **Bulk**, **Other**, and **Pets** recorded relatively low performance compared to major grocery departments. Retailer should evaluate product assortment, pricing strategies, and promotional campaigns for these categories to improve visibility and stimulate customer demand.

Overall Recommendation

The analysis indicates that business growth can be achieved by focusing on **high-demand essential grocery products**, maintaining product availability, strengthening customer loyalty initiatives, implementing personalized marketing strategies, and optimizing inventory based on customer purchasing patterns. These initiatives can improve customer satisfaction, increase repeat purchases, and support long-term business performance.

Future Scope

Future work can include:

- Sales forecasting using machine learning.
- Customer segmentation using RFM analysis or clustering.
- Market basket analysis to identify frequently purchased product combinations.
- Demand prediction for inventory optimization.
- Customer lifetime value analysis.

Conclusion

This project analysed a retail dataset to understand customer purchasing behavior, product performance, and sales trends using **MySQL** and **Power BI**. Data quality issues in the Products table were identified and corrected using Excel before conducting the analysis, to ensure that we get accurate and reliable results.

This analysis revealed that **Produce, Dairy & Eggs**, and **Beverages** as the most popular departments, with products such as **Bananas, Organic Whole Milk**, and **Organic Strawberries** having highest customer demand and reorder rates. Customer purchasing behavior indicated regular shopping patterns, moderate basket sizes, and strong repeat purchasing of essential grocery items. Peak shopping activity was observed during the **afternoon hours**, while specific days of the week recorded higher order volumes and larger basket sizes.

The findings show that the customer demand is primarily driven by essential grocery products rather than the size of the product assortment. Based on the insights, recommendations such as optimizing inventory, implementing personalized marketing, strengthening loyalty programs, and promoting high-demand products can help improve customer satisfaction, increase repeat purchases, and support better business decision-making.

Overall, this project demonstrates how data analytics techniques can transform raw retail data into meaningful business insights that support operational efficiency and strategic planning.